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Questions

Q1.

(a) Show that the equation

$$8 \cos \theta = 3 \operatorname{cosec} \theta$$

can be written in the form

$$\sin 2\theta = k$$

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where k is a constant to be found.

(3)

(b) Hence find the smallest positive solution of the equation

$$8 \cos \theta = 3 \operatorname{cosec} \theta$$

giving your answer, in degrees, to one decimal place.

(2)

(Total for question = 5 marks)

(Q02 WMA13/01, Jan 2022)

Q2.

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In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < x \leq \pi$, the equation

$$2 \sec^2 x - 3 \tan x = 2$$

giving the answers, as appropriate, to 3 significant figures.

(4)

(ii) Prove that

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$$\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} \equiv 2$$

(4)

(Total for question = 8 marks)

(Q09 WMA13/01, Jan 2022)

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Q3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Given that $\cos 2\theta - \sin 3\theta \neq 0$

(a) prove that

$$\frac{\cos^2 \theta}{\cos 2\theta - \sin 3\theta} \equiv \frac{1 + \sin \theta}{1 - 2\sin \theta - 4\sin^2 \theta}$$

(4)

(b) Hence solve, for $0 < \theta \leqslant 360°$

$$\frac{\cos^2 \theta}{\cos 2\theta - \sin 3\theta} = 2\operatorname{cosec} \theta$$

Give your answers to one decimal place.

(5)

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(Total for question = 9 marks)

(Q09 WMA13/01, Oct 2022)

Q4.

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$f(\theta) = 5\cos\theta - 4\sin\theta \qquad \theta \in \mathbb{R}$

(a) Express f(θ) in the form $R\cos(\theta + \alpha)$, where R and α are constants, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

Give the exact value of R and give the value of α, in radians, to 3 decimal places.

(3)

The curve with equation $y = \cos\theta$ is transformed onto the curve with equation $y = f(\theta)$ by a sequence of two transformations.

Given that the first transformation is a stretch and the second a translation,

(b) (i) describe fully the transformation that is a stretch,

(ii) describe fully the transformation that is a translation.

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(2)

Given

$$g(\theta) = \frac{90}{4 + (f(\theta))^2} \qquad \theta \in \mathbb{R}$$

(c) find the range of g.

(2)

(Total for question = 7 marks)

(**Q09 WMA13/01, Jan 2020**)

Q5.

(a) Use the substitution $t = \tan x$ to show that the equation

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$$12\tan 2x + 5\cot x \sec^2 x = 0$$

can be written in the form

$$5t^4 - 24t^2 - 5 = 0$$

(4)

(b) Hence solve, for $0 \leq x < 360°$, the equation

$$12\tan 2x + 5\cot x \sec^2 x = 0$$

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Show each stage of your working and give your answers to one decimal place.

(4)

(Total for question = 8 marks)

(Q05 WMA13/01, Jan 2020)

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Q6.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\frac{\cos 2x}{\sin x} + \frac{\sin 2x}{\cos x} \equiv \operatorname{cosec} x \qquad x \neq \frac{n\pi}{2} \qquad n \in \mathbb{Z}$$

(3)

(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$

$$\left(\frac{\cos 2\theta}{\sin \theta} + \frac{\sin 2\theta}{\cos \theta}\right)^2 = 6\cot\theta - 4$$

giving your answers to 3 significant figures as appropriate.

(5)

(c) Using the result from part (a), or otherwise, find the exact value of

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(\frac{\cos 2x}{\sin x} + \frac{\sin 2x}{\cos x}\right) \cot x \, dx$$

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(2)

(Total for question = 10 marks)

(Q09 WMA13/01, June 2023)

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Q7.

(a) Prove that

$$\sec 2A + \tan 2A \equiv \frac{\cos A + \sin A}{\cos A - \sin A} \qquad A \neq \frac{(2n+1)\pi}{4} \qquad n \in \mathbb{Z}$$

(5)

(b) Hence solve, for $0 \leq \theta < 2\pi$

$$\sec 2\theta + \tan 2\theta = \frac{1}{2}$$

Give your answers to 3 decimal places.

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(4)

(Total for question = 9 marks)

(**Q09 WMA13/01, Specimen papers**)

Q8.

(a) Show that

$$\sin 3x \equiv 3\sin x - 4\sin^3 x$$

(4)

(b) Hence find, using algebraic integration,

$$\int_0^{\frac{\pi}{3}} \sin^3 x \, dx$$

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(4)

(Total for question = 8 marks)

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(Q05 WMA13/01, Oct 2020)

Q9.

(a) Express $12\sin x - 5\cos x$ in the form $R\sin(x - \alpha)$, where R and α are constants, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give the exact value of R and give the value of α in radians, to 3 decimal places.

(3)

The function g is defined by

$$g(\theta) = 10 + 12\sin\left(2\theta - \frac{\pi}{6}\right) - 5\cos\left(2\theta - \frac{\pi}{6}\right) \qquad \theta > 0$$

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Find

(b) (i) the minimum value of $g(\theta)$

(ii) the smallest value of θ at which the minimum value occurs.

(3)

The function h is defined by

$$h(\beta) = 10 - (12\sin\beta - 5\cos\beta)^2$$

(c) Find the range of h.

(2)

(Total for question = 8 marks)

(Q09 WMA13/01, June 2021)

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Q10.

(a) Show that

$$\frac{1 - \cos 2x}{2\sin 2x} \equiv k\tan x \qquad x \neq (90n)^\circ \qquad n \in \mathbb{Z}$$

where k is a constant to be found.

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(3)

(b) Hence solve, for $0 < \theta < 90°$

$$\frac{9(1-\cos 2\theta)}{2\sin 2\theta} = 2\sec^2\theta$$

giving your answers to one decimal place.

(*Solutions based entirely on graphical or numerical methods are not acceptable.*)

(6)

(Total for question = 9 marks)

(**Q02 WMA13/01, June 2021**)

Q11.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Prove that

$$\frac{\sin 2x}{\cos x} + \frac{\cos 2x}{\sin x} \equiv \operatorname{cosec} x \qquad x \neq \frac{n\pi}{2} \quad n \in \mathbb{Z}$$

(3)

(b) Hence solve, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$7 + \frac{\sin 4\theta}{\cos 2\theta} + \frac{\cos 4\theta}{\sin 2\theta} = 3\cot^2 2\theta$$

giving your answers in radians to 3 significant figures where appropriate.

(6)

(Total for question = 9 marks)

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(Q07 WMA13/01, Jan 2021)

Q12.

Solve, for $0 \leq x < 360°$, the equation

$$2\cos 2x = 7\cos x$$

giving your solution to one decimal place.

(*Solutions based entirely on graphical or numerical methods are not acceptable.*)

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(Total for question = 5 marks)

(Q01 WMA13/01, Oct 2020)

Q13.

(a) Prove that

$$2\,\text{cosec}^2 2\theta\,(1 - \cos 2\theta) \equiv 1 + \tan^2\theta$$

(4)

(b) Hence solve for $0 < x < 360°$, where $x \neq (90n)°$, $n \in \mathbb{N}$, the equation

$$2\,\text{cosec}^2 2x\,(1 - \cos 2x) = 4 + 3\sec x$$

giving your answers to one decimal place.

(*Solutions relying entirely on calculator technology are not acceptable.*)

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(4)

(Total for question = 8 marks)

(Q08 WMA13/01, Oct 2023)

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Q14.

In this question you should show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2 \sin (\theta - 30°) = 5 \cos \theta$$

can be written in the form

$$\tan \theta = 2\sqrt{3}$$

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(4)

(b) Hence, or otherwise, solve for $0 \leq x \leq 360°$

$$2 \sin (x - 10°) = 5 \cos (x + 20°)$$

giving your answers to one decimal place.

(3)

(Total for question = 7 marks)

(**Q04 WMA13/01, Oct 2021**)

Q15.

In this question you must show all stages of your working.

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Solutions relying entirely on calculator technology are not acceptable.

(a) Prove that

$$\cot^2 x - \tan^2 x \equiv 4 \cot 2x \operatorname{cosec} 2x \qquad x \neq \frac{n\pi}{2} \quad n \in \mathbb{Z}$$

(4)

(b) Hence solve, for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

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$$4\cot 2\theta \operatorname{cosec} 2\theta = 2\tan^2\theta$$

giving your answers to 2 decimal places.

(5)

(Total for question = 9 marks)

(**Q05 WMA13/01, Jan 2023**)

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Q16.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < x < \pi$

$$(x-2)\left(\sqrt{3}\sec x + 2\right) = 0$$

(3)

(ii) Solve, for $0 < \theta < 360°$

$$10\sin\theta = 3\cos 2\theta$$

(4)

(Total for question = 7 marks)

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(**Q05 WMA13/01, June 2023**)

Q17.

$$f(x) = \cos x + 2\sin x$$

(a) Express f(x) in the form $R\cos(x-\alpha)$, where R and α are constants,

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$R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and give the value of α, in radians, to 3 decimal places.

(3)

$$g(x) = 3 - 7f(2x)$$

(b) Using the answer to part (a),

(i) write down the exact maximum value of g(x),

(ii) find the smallest positive value of x for which this maximum value occurs, giving your answer
to 2 decimal places.

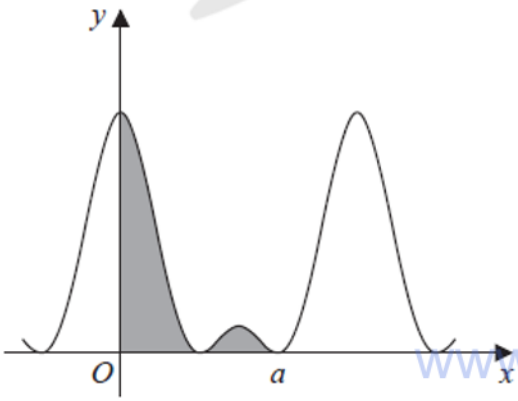
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(3)

(Total for question = 6 marks)

(**Q02 WMA13/01, Jan 2023**)

Q18.



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Figure 4

Figure 4 shows a sketch of part of the curve with equation

$$y = (1 + 2\cos 2x)^2$$

(a) Show that

$$(1 + 2\cos 2x)^2 \equiv p + q\cos 2x + r\cos 4x$$

where p, q and r are constants to be found.

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(2)

The curve touches the positive x-axis for the second time when $x = a$, as shown in Figure 4.

The regions bounded by the curve, the y-axis and the x-axis up to $x = a$ are shown shaded in Figure 4.

(b) Find, using algebraic integration and making your method clear, the exact total area of the shaded regions. Write your answer in simplest form.

(5)

(Total for question = 7 marks)

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(**Q10 WMA13/01, Oct 2021**)

Q19.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Express $8 \sin x - 15 \cos x$ in the form $R \sin (x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R, and give the value of α, in radians, to 4 significant figures.

(3)

$$f(x) = \frac{15}{41 + 16\sin x - 30\cos x} \qquad x > 0$$

(b) Find

(i) the minimum value of f(x)

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(ii) the smallest value of x at which this minimum value occurs.

(4)

(c) State the y coordinate of the minimum points on the curve with equation

$$y = 2f(x) - 5 \qquad x > 0$$

(1)

(d) State the smallest value of x at which a maximum point occurs for the curve with equation

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$y = -f(2x) \quad x > 0$

(1)

(Total for question = 9 marks)

(**Q08 WMA13/01, Oct 2022**)

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Q20.

(a) Using the identity for $\cos(A + B)$, prove that

$$\cos 2A \equiv 2\cos^2 A - 1$$

(2)

(b) Hence, using algebraic integration, find the exact value of

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{8}} (5 - 4\cos^2 3x)\,dx$$

(4)

(Total for question = 6 marks)

(**Q03 WMA13/01, Oct 2023**)

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Q21.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2\sin\theta\,(3\cot^2 2\theta - 7) = 13\sec\theta$$

can be written as

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$3\text{cosec}^2 2\theta - 13\text{cosec}2\theta - 10 = 0$

(4)

(b) Hence solve, for $0 < \theta < \frac{\pi}{2}$, the equation

$$2\sin\theta\,(3\cot^2 2\theta - 7) = 13\sec\theta$$

giving your answers to 3 significant figures.

(4)

(Total for question = 8 marks)

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(Q07 WMA13/01, June 2022)

Q22.

(a) Express $\cos x + 4\sin x$ in the form $R\cos(x - \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$

Give the exact value of R and give the value of α, in radians, to 3 decimal places.

(3)

A scientist is studying the behaviour of seabirds in a colony.

She models the height above sea level, H metres, of one of the birds in the colony by the equation

$$H = \frac{24}{3 + \cos\left(\frac{1}{2}t\right) + 4\sin\left(\frac{1}{2}t\right)} \qquad 0 \leqslant t \leqslant 6.5$$

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where t seconds is the time after it leaves the nest.

Find, according to the model,

(b) the minimum height of the seabird above sea level, giving your answer to the nearest cm,

(2)

(c) the value of t, to 2 decimal places, when $H = 10$

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(4)

(Total for question = 9 marks)

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